zomato

REVENUE & OPERATIONS ANALYTICS

Restaurant Performance • Customer Insights • Delivery Operations

Presented by

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TOOLS

SQL • Python • Power BI • Excel

DATASETS

6 Tables • 800K+ Records

FOCUS

Revenue • Delivery • Cuisine Trends

Analyzing Zomato's restaurant, customer, menu, order, and delivery ecosystems to drive data-backed decisions

Project Overview

Business Problem • Objectives • Expected Outcomes

BUSINESS PROBLEM

Zomato generates vast amounts of transactional, operational, and customer data across hundreds of cities. Without structured analysis, critical insights on revenue drivers, delivery efficiency, and customer preferences remain hidden — preventing strategic decision-making.

PROJECT OBJECTIVES

1. Analyze revenue performance across cities and time periods
2. Understand customer demographics, behavior, and ordering patterns
3. Evaluate restaurant distribution, cuisine trends, and ratings
4. Assess delivery operations: traffic, weather, and time efficiency
5. Build an interactive Power BI dashboard for stakeholder use

EXPECTED OUTCOMES

- 💡 Actionable business insights from 800K+ records
- 📊 Interactive multi-page Power BI dashboard
- 🔍 City-wise & cuisine-wise performance report
- 🚚 Delivery optimization recommendations
- 🎯 Data-driven marketing & expansion strategy

Dataset Overview

6 Structured Tables • 800K+ Total Records • Relational Schema

TOTAL RECORDS ≈ 800K+

USERS

RECORDS

100,000

Key Columns: user_id, name, age, gender, occupation, income, education

ORDERS

RECORDS

150,281

Key Columns: order_date, sales_qty, sales_amount, currency, user_id, r_id

RESTAURANT

RECORDS

148,544

Key Columns: id, name, city, rating, cost, cuisine, address

FOOD

RECORDS

371,641

Key Columns: f_id, item, veg_or_non_veg

MENU

RECORDS

Linked

Key Columns: restaurant_id, food_id, price, portion_size

DELIVERY OPS

RECORDS

43,853

Key Columns: age, rating, weather, traffic, vehicle, time_taken

Database Design & ER Diagram

Entity Relationships • Normalized Schema • Primary & Foreign Keys

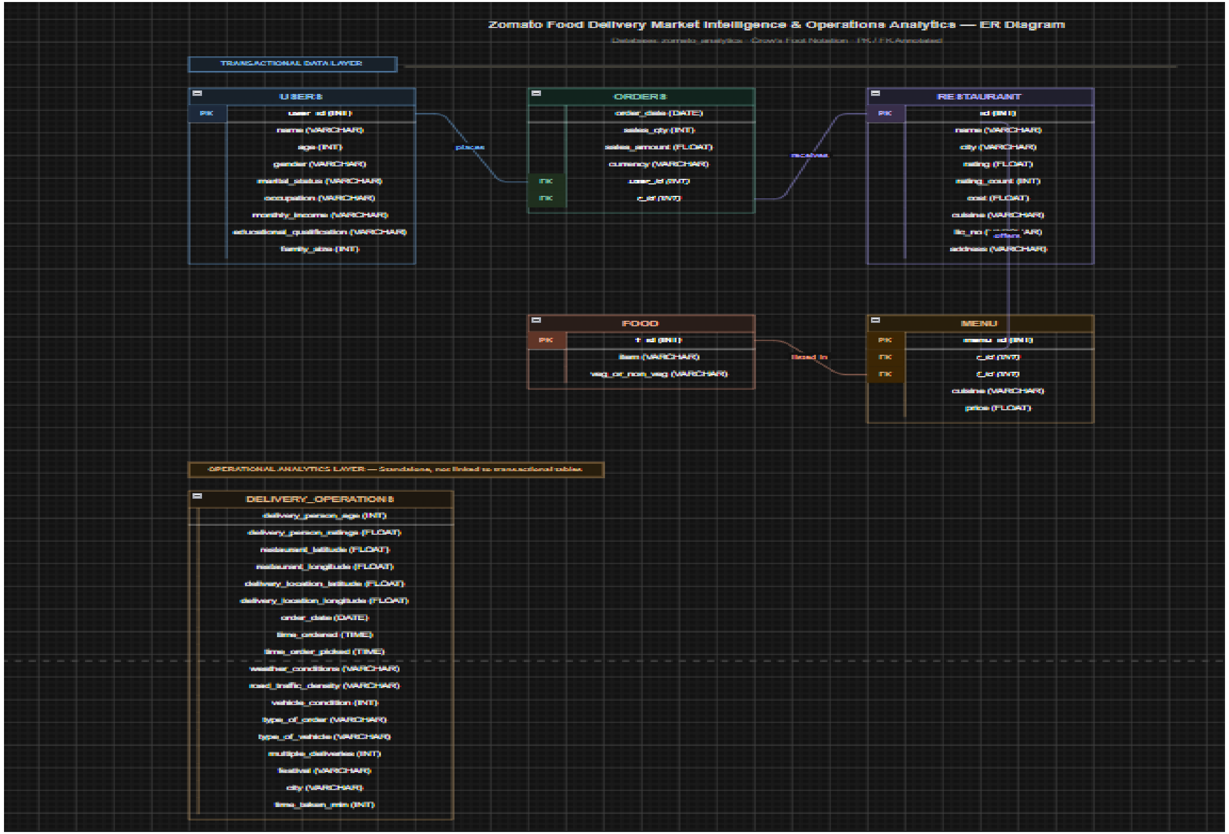
WHY ER MODELING?

- ✦ Prevents data redundancy across large tables
- ✦ Enables efficient multi-table JOIN queries
- ✦ Maintains referential integrity across entities
- ✦ Supports scalable analytics and reporting

KEY RELATIONSHIPS

- → Users → Orders (user_id)
- → Restaurants → Orders (r_id)
- → Restaurants → Menu (r_id)
- → Food → Menu (f_id)
- → Menu → Price & Portions

ER Diagram



Zomato Food Delivery Market Intelligence & Operations Analytics — ER Diagram
TRANSACTIONAL DATA LAYER
USERS
PK
name (VARCHAR)
age (INT)
gender (VARCHAR)
marital_status (VARCHAR)
occupation (VARCHAR)
monthly_income (VARCHAR)
educational_qualification (VARCHAR)
family_size (INT)
places
ORDERS
order_date (DATE)
sales_qty (INT)
sales_amount (FLOAT)
currency (VARCHAR)
FK
FK
receives
RESTAURANT
PK
name (VARCHAR)
city (VARCHAR)
rating (FLOAT)
rating_count (INT)
cost (FLOAT)
cuisine (VARCHAR)
address (VARCHAR)
offers
FOOD
PK
item (VARCHAR)
veg_or_non_veg (VARCHAR)
listed in
MENU
PK
FK
FK
cuisine (VARCHAR)
price (FLOAT)
DELIVERY_OPERATIONS
delivery_person_age (INT)
delivery_person_ratings (FLOAT)
restaurant_latitude (FLOAT)
restaurant_longitude (FLOAT)
delivery_location_latitude (FLOAT)
delivery_location_longitude (FLOAT)
order_date (DATE)
time_ordered (TIME)
time_order_picked (TIME)
weather_conditions (VARCHAR)
road_traffic_density (VARCHAR)
vehicle_condition (INT)
type_of_order (VARCHAR)
type_of_vehicle (VARCHAR)
multiple_deliveries (INT)
festival (VARCHAR)
city (VARCHAR)
time_taken_min (INT)

Data Cleaning Process

Ensuring Data Quality Before Analysis

01

Duplicate Removal

Identified and dropped duplicate records using SQL DISTINCT and Python drop_duplicates() to ensure accurate aggregations.

02

Missing Value Handling

Replaced NULL entries with mean/median for numeric fields; flagged missing categorical values as 'Unknown' for restaurant ratings and menu data.

03

Data Type Corrections

Converted order_date to DATE type, sales_amount to FLOAT, and ensured rating fields were stored as numeric for comparison operations.

04

Menu Table Cleaning

Parsed nested JSON menu references, standardized cuisine categories, and resolved encoding issues in food item names.

05

SQL Optimization

Created indexed views and optimized JOIN conditions to improve query performance across 800K+ records during analysis.

SQL Analysis

Core Business Queries Across 5 Analysis Dimensions

Revenue Analysis

- Total revenue by month and city
- Average order value per customer
- Monthly revenue trend identification

```sql
SELECT
    MONTH(order_date) AS month_no,
    COUNT(*) AS total_orders
FROM orders
GROUP BY MONTH(order_date)
ORDER BY month_no;
```

Customer Analysis

- Customer demographics breakdown
- High-value customer segmentation
- Order frequency per user

```sql
SELECT
    u.occupation,
    COUNT(*) AS total_orders
FROM users u
JOIN orders o
    ON u.user_id = o.user_id
GROUP BY u.occupation
ORDER BY total_orders DESC;
```

Restaurant Analysis

- Top-rated restaurants per city
- Restaurant count by city density
- Cost vs. rating correlation

```sql
SELECT
    city,
    ROUND(AVG(cost),2) AS avg_cost
FROM restaurant
GROUP BY city
ORDER BY avg_cost DESC;
```

Cuisine Analysis

- Most popular cuisines by order volume
- Cuisine distribution across regions
- Veg vs. Non-veg preference

```sql
SELECT
    cuisine,
    COUNT(*) AS total_restaurants
FROM restaurant
GROUP BY cuisine
ORDER BY total_restaurants DESC;
```

Delivery Analysis

- Average delivery time by traffic density
- Weather impact on delivery duration
- Vehicle type performance

```sql
SELECT
    Road_traffic_density,
    ROUND(AVG(`Time_taken (min)`),2) AS avg_delivery_time
FROM delivery_operations
GROUP BY Road_traffic_density
ORDER BY avg_delivery_time DESC;
```

Advanced SQL Queries

Joins • Aggregations • Window Functions • Business KPIs

Multi-Table JOINs

Integrated customer, order, and restaurant data to generate cross-functional business insights.

```
SELECT
    u.gender,
    COUNT(*) AS total_orders
FROM users u
JOIN orders o
    ON u.user_id = o.user_id
GROUP BY u.gender;
```

Window Functions

Use ROW_NUMBER() and RANK() to identify top-performing restaurants within each city.

```
SELECT *
FROM (
    SELECT
        city,
        name,
        rating,
        ROW_NUMBER() OVER(
            PARTITION BY city
            ORDER BY rating DESC
        ) AS rn
    FROM restaurant
) x
WHERE rn <= 5;
```

KPI Analytics & Aggregation

Generated business KPIs including Total Revenue, Avg Order Value, Customer Lifetime Value, and Delivery Efficiency Score.

```
SELECT
    Weather_conditions,
    ROUND(AVG(`Time_taken (min)`),2) AS avg_delivery_time
FROM delivery_operations
GROUP BY Weather_conditions
ORDER BY avg_delivery_time DESC;
```

BUSINESS KPIs GENERATED

- **Total Revenue**
- **Avg Order Value**
- **Customer LTV**
- **Top City by Orders**
- **Avg Delivery Time**
- **Top Cuisine by Revenue**
- **Restaurant Rating Avg**
- **Traffic Impact Score**

Python Analytics

Exploratory Data Analysis with Pandas & Matplotlib

```python
orders['year'] = pd.to_datetime(
    orders['order_date']
).dt.year

rev = orders.groupby(
    'year'
)['sales_amount'].sum()

print(rev)
```

```
year
2017.0     93568402.0
2018.0    414308941.0
2019.0    336452114.0
2020.0    142235559.0
Name: sales_amount, dtype: float64
```

```python
# Top Cities

top_cities = restaurant["city"].value_counts().head(10)

top_cities.plot(kind="bar")
plt.title("Top 10 Cities by Restaurant Count")
plt.show()
```

Revenue Trend Analysis

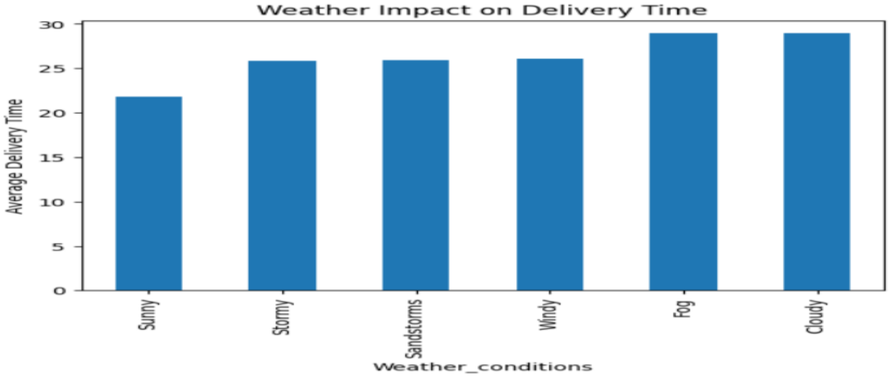
Plotted year-over-year revenue (2017–2020) to identify peak periods and seasonal patterns in ordering behavior

Cuisine Distribution

Bar charts and pie charts showing cuisine popularity across top cities to identify regional preferences.

```python
#Weather Impact
weather_analysis = (
    delivery.groupby("Weather_conditions")
    ["Time_taken (min)"]
    .mean()
    .sort_values()
)

print(weather_analysis)

weather_analysis.plot(kind="bar")

plt.title("Weather Impact on Delivery Time")
plt.ylabel("Average Delivery Time")

plt.show()
```

```python
highest_rated_city = restaurant.groupby(
    "city"
)["rating"]\
.mean()\
.sort_values(
    ascending=False
).head(10)
```




```
city
Chopda                        4.253846
Kumta                         4.200000
Kadayanallur                  4.175000
Lulu Mall,Kochi               4.163636
South-goa                     4.142857
Mylapore,Chennai              4.138065
Dhanbad                       4.133333
Kidderpore,Kolkata            4.125926
South Kolkata,Kolkata         4.120884
Adyar,Chennai                 4.117102
```

🚚 Delivery Performance

Analyzed average delivery time by traffic density and weather conditions using grouped statistics.

👥 Customer Insights

Demographic segmentation by age, income, occupation, and gender to profile Zomato's key customer segments.

Power BI Dashboard Development

End-to-End BI Workflow: Import → Model → DAX → Visualize

1	2	3	4	5
Data Import	Data Modeling	Relationships	DAX Measures	Dashboard Design

DAX MEASURES CREATED

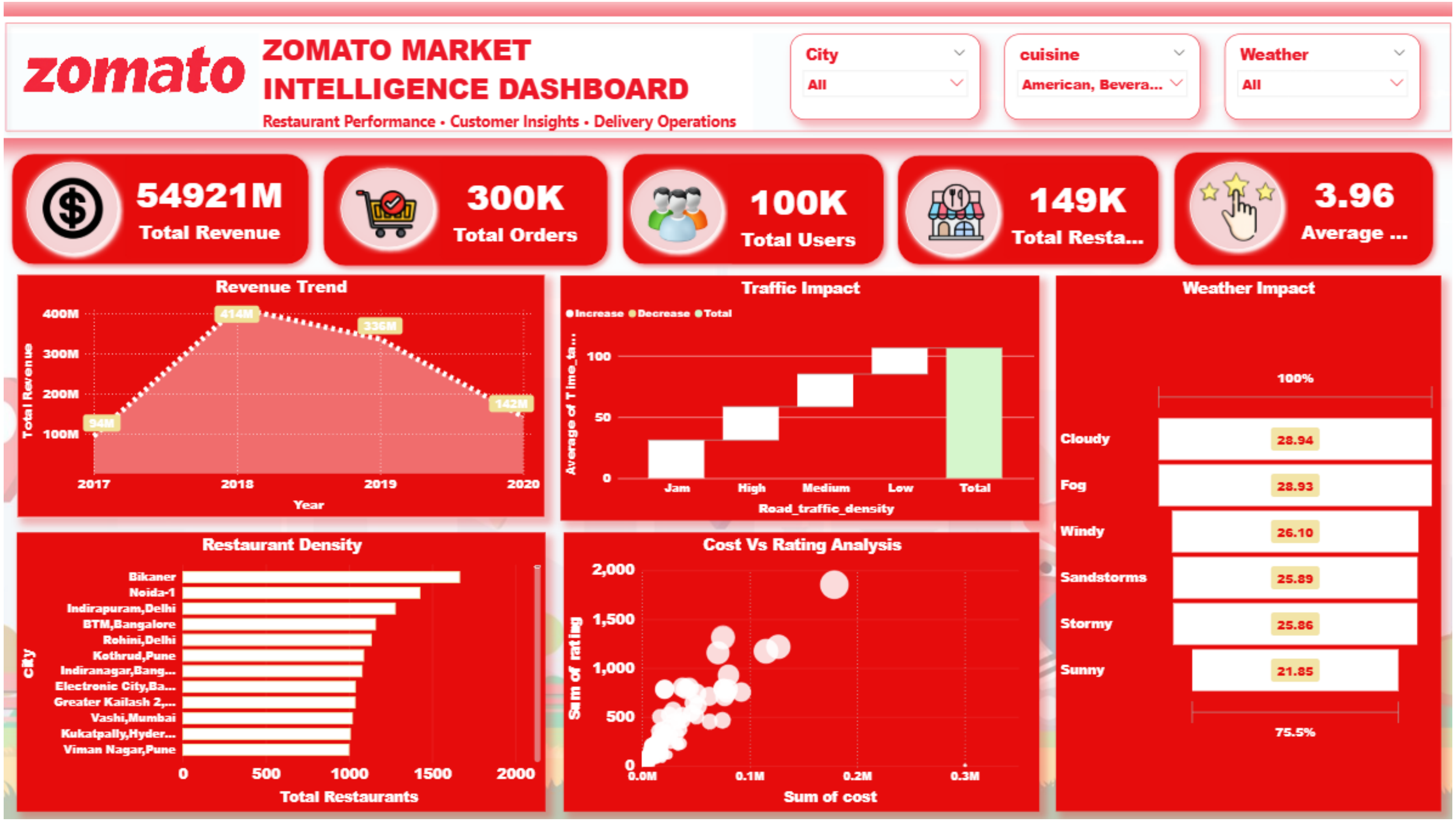
Total Revenue	`= SUM(orders[sales_amount])`
Avg Order Value	`= AVERAGE(orders[sales_amount])`
Total Orders	`= COUNTROWS(orders)`
Avg Delivery Time	`= AVERAGE(delivery[Time_taken (min)])`
Avg Rating	`= AVERAGE(restaurant[rating])`

INTERACTIVE FEATURES

⚡ **City Slicer**	Filter all visuals by selected city
⚡ **Cuisine Filter**	Drill down into specific cuisine types
⚡ **Weather Dropdown**	Analyze impact by weather condition
⚡ **Cross-filtering**	Click any chart to filter related visuals
⚡ **Drill-through**	Navigate from summary to detail level

Dashboard Overview

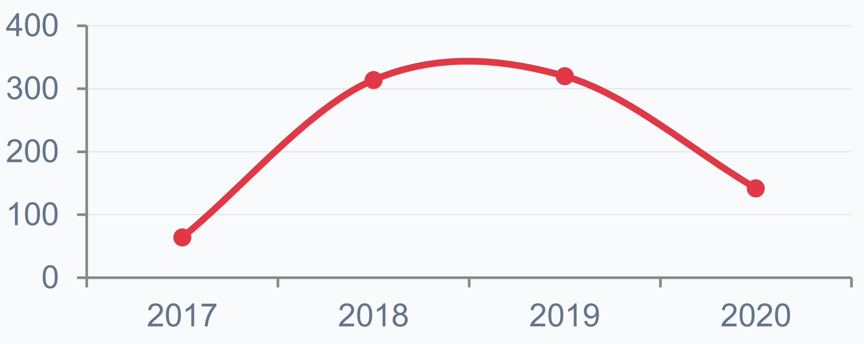
Zomato Market Intelligence Dashboard — Power BI



Key Visualizations

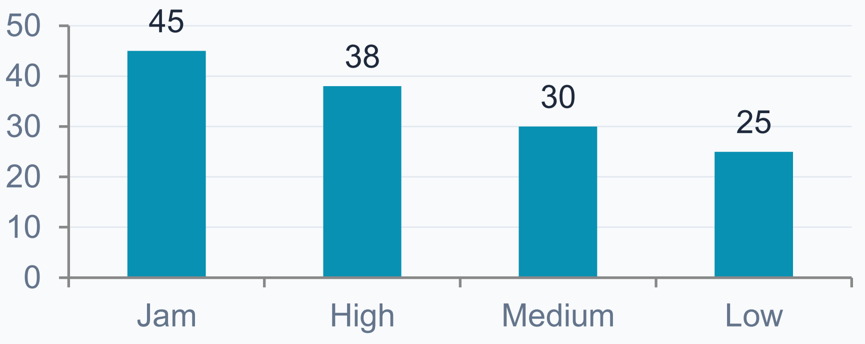
5 Core Charts Driving Business Intelligence

📈 Revenue Trend (2017–2020)



Revenue peaked in 2018 then declined through 2020, suggesting pandemic-era impact on food delivery patterns.

🚦 Traffic Impact on Delivery



Jam traffic conditions result in the highest average delivery time, nearly 2x compared to low-traffic routes.

🌧️ Weather Impact

Cloudy & Foggy conditions have highest avg delivery time (~29 min). Sunny conditions = fastest deliveries (~22 min).

🏙️ Restaurant Density

Bikaner, Noida-1, and Indirapuram lead in restaurant count. Metro cities like Bangalore and Delhi show high market saturation.

⭐ Cost vs Rating Analysis

Higher-cost restaurants generally trend toward higher ratings, though premium outliers exist with surprisingly low scores.

Key Business Insights

Data-Driven Findings Across All Analysis Dimensions

01 City Dominance

Bikaner, Noida-1, and Indirapuram lead in restaurant density. Metros like Bangalore and Delhi show market saturation, signaling opportunity in tier-2 cities.

02 Weather & Delivery

Cloudy and Foggy conditions add ~7 minutes to average delivery time vs. Sunny conditions (~22 min). Weather is a significant operational risk factor.

03 Cost-Rating Correlation

Higher-cost restaurants trend toward better ratings, but the relationship is not linear — value restaurants with great service can outperform premium ones.

04 Traffic & Performance

Jam traffic density results in ~80% longer delivery time compared to low-traffic routes. Route optimization is critical for on-time delivery performance.

05 Regional Cuisine Trends

Cuisine preferences vary significantly by region. Local and regional cuisines dominate tier-2 cities while fast food and beverages lead in metro areas.

06 Revenue Peak (2018)

Revenue peaked in 2018 and declined through 2020. Year-over-year analysis reveals the need for customer retention programs to sustain growth trajectories.

Business Recommendations

Actionable Strategies Based on Data-Driven Insights

Optimize Delivery in Adverse Weather

Deploy dynamic surge pricing and pre-position delivery partners near high-demand zones during cloudy/foggy conditions to maintain delivery SLAs.

IMPACT: HIGH

Expand into High-Potential Cities

Focus restaurant onboarding in tier-2 cities like Bikaner and Noida where restaurant density is high but delivery infrastructure is still developing.

IMPACT: HIGH

Promote High-Rated Restaurants

Feature top-rated restaurants prominently in app recommendations and targeted email campaigns to increase average order value and repeat orders.

IMPACT: MEDIUM

Traffic-Aware Delivery Allocation

Implement real-time traffic data integration to assign delivery orders to partners on low-congestion routes, reducing average delivery time by an estimated 20-30%.

IMPACT: HIGH

Cuisine-Targeted Marketing

Use cuisine trend data to create geo-targeted promotions — push regional specialties in tier-2 cities and fast food combos in metro areas during peak hours.

IMPACT: MEDIUM

CONCLUSION

📋 **Project Summary**

Analyzed 800K+ records across 6 Zomato datasets using SQL, Python, and Power BI to deliver a comprehensive Market Intelligence Dashboard covering revenue, delivery, cuisine, and customer insights.

💼 **Business Value**

Delivered 6 key business insights and 5 actionable recommendations to optimize delivery operations, support restaurant expansion decisions, and drive targeted marketing strategies.

🎓 **Learning Outcomes**

Gained hands-on expertise in end-to-end analytics: data modeling, advanced SQL, Python EDA, DAX measures, and interactive dashboard design — all applied to real-world business scenarios.

🚀 **Future Improvements**

Planned enhancements include machine learning for delivery time prediction, NLP-based customer review sentiment analysis, and real-time data refresh via Zomato API integration.

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Market Intelligence Dashboard

- SQL + Advanced SQL
- Python (Pandas, Matplotlib)
- Power BI + DAX
- Excel (Data Prep)

Thank You

By Taniya